Resolution of the Linear MDS Conjecture over Finite Fields of Even Order

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Abstract

Maximum distance separable (MDS) codes give the best possible protection against erasures: they store k data symbols as N symbols so that *any* k of the N recover the data. They safeguard RAID arrays, cloud storage, QR codes and the AES cipher, and in practice they are built over fields of even order such as the byte field $\mathbb{F}_{256}$. How long can such a code be? The MDS conjecture, which goes back to a 1955 question of Beniamino Segre, predicts that a k-dimensional linear MDS code over $\mathbb{F}_q$ with $2 \le k \le q$ has length at most $q + 1$, except for length $q + 2$ when q is even and $k \in \{3, q - 1\}$. In 2012 Simeon Ball proved the conjecture for every prime field, but his method gives nothing new in characteristic two, where the conjecture had remained open for more than seventy years outside restricted ranges of k.

This report resolves the linear MDS conjecture for every finite field of even order and every dimension $2 \le k \le q$. Duality reduces the problem to excluding rank-r arcs of length $q + 2$ for $4 \le r \le q/2 + 1$. We introduce a common-height method: contracting $t = r - 3$ vectors of a hypothetical arc in a single coordinate system yields one plane arc together with t lifts into space, and these lifts and three coordinate rows form $t + 3$ linearly independent vectors. Using characteristic-two identities for derivatives, we prove that every lift satisfies one fixed linear system, whose solutions inject, modulo a three-dimensional space of shears, into a space of degree-$(t - 2)$ annihilators; a rank induction bounds the dimension of that space by $t - 1$. All lifts therefore lie in one space of dimension at most $t + 2$, exactly one dimension too small. The complete proof has been formally verified in the Lean 4 proof assistant. In addition, every lemma was tested by computer on 250 explicit plane arcs over $\mathbb{F}_8$, $\mathbb{F}_{16}$, $\mathbb{F}_{32}$ and $\mathbb{F}_{64}$ without a single failure, and exhaustive search confirmed the exact maximum lengths over $\mathbb{F}_2$, $\mathbb{F}_4$ and $\mathbb{F}_8$. As a consequence, extended Reed–Solomon codes over $\mathbb{F}_{256}$ have the largest possible length in every dimension $4 \le k \le 254$.

1 Introduction

1.1 MDS codes and where they are used

Suppose a file is cut into k pieces and stored on $N > k$ disks so that it survives the failure of any $N - k$ disks. If every disk holds one symbol from a finite alphabet, the best one can hope for is that *any* k of the N disks rebuild the file, since k unknown symbols cannot be recovered from fewer than k symbols. Linear codes with this property are called *maximum distance separable* (MDS). Error-correcting codes go back to Shannon and Hamming [64, 30]. A k-dimensional linear code of length N is MDS when its minimum Hamming distance equals $N - k + 1$, the largest value allowed by the Singleton bound [65, 47, 37]. The smallest interesting example stores two symbols a, b as $a, b, a + b$ (Figure 1).

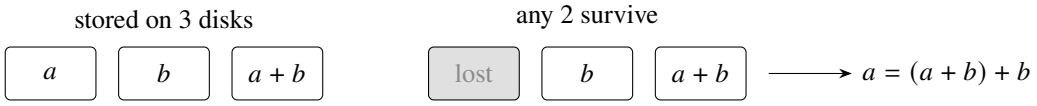


Figure 1: The smallest MDS code ($k = 2$, $N = 3$) stores a, b as $a, b, a + b$. Any two of the three stored symbols determine a and b; in characteristic two, subtraction is the same as addition.

Image created by the student researcher using TikZ [70], 2026.

MDS codes protect disk arrays (RAID) [52]; RAID-6 uses Reed–Solomon (RS) codes [56, 53, 54], the standard MDS family, or XOR-based MDS array codes such as EVENODD [8]. RS and other MDS codes are also used in erasure-coded distributed storage [21, 36], in packet-loss recovery for network protocols [57], in compact discs and QR codes [73, 39], and in threshold secret sharing [63, 49]. In cryptography, the MixColumns step of the AES block cipher [51] multiplies each column of the state by an MDS matrix over $\mathbb{F}_{2^8}$, so that every byte of a column influences every byte of the output column [19, Chs. 3, 9], [40]. In these systems the alphabet is typically a finite field of *even* order $q = 2^m$, most often $\mathbb{F}_{256}$, whose elements are bytes and whose addition is bitwise XOR [44, 53, 57, 51].

This raises a basic engineering limit. For a fixed field size q and dimension k, how long can an MDS code be? Extended RS codes have length $q + 1$ [47, 58]. If longer MDS codes existed, a storage system over $\mathbb{F}_{256}$ could, for example, protect more than 257 disks with the MDS guarantee without moving to a larger field. The result of this report shows that, over every field of even order, longer codes do not exist except in the two classical exceptional dimensions. Engineers who need more than $q + 1$ symbols must therefore enlarge the field or give up the MDS property, as locally repairable codes do [27, 36].

1.2 The MDS conjecture

The columns of a generator matrix of an MDS code form an *arc*: a list of vectors in K^k, every k of which are linearly independent (Lemma 2.3). Arcs were studied by Bose [10] and Bush [13] in connection with

statistical designs, and Segre [59, 60] made the determination of their maximum size a central problem of finite geometry [33, 35]. The predicted answer is now known as the MDS conjecture [47, 34, 3, 7].

MDS Conjecture. *Let $2 \le k \le q$. A k-dimensional linear MDS code over $\mathbb{F}_q$ has length at most $q+1$, except that the maximum is $q+2$ when q is even and $k \in \{3, q-1\}$.*

For $k > q$ the maximum length is $k+1$ [13]. Ball proved the conjecture for every prime q [3], and Ball and De Beule extended the proof to $q = p^h$ with $k \le 2p-2$ [5]; for $p = 2$ this range contains no new cases, and neither do later refinements [17, 6]. For even q the known cases were $k = 4$ and $k = 5$ [14], $k = 6$ [16], $k = 7$ for $q \ge 16$ [48], and $k < \frac{1}{2}\sqrt{q} + \frac{15}{4}$ [67], improving earlier bounds of Bruen, Thas and Blokhuis [11, 9], together with the dual values $q+2-k$. The conjecture was also known for all $q \le 27$ [34, 3], and computer classifications determined all arcs for $4 \le q \le 19$ [41]. For large even q the middle range of k remained open; see the survey of Ball and Lavrauw [7].

1.3 The result and the approach

Theorem 1.1 (Main Theorem)**.** *Let K be a finite field of even order q and let $2 \le k \le q$. Put*

$$B(q,k) = \begin{cases} q+2, & k = 3 \text{ or } k = q-1, \\ q+1, & \text{otherwise.} \end{cases}$$

Every k-dimensional linear MDS code over K has length at most $B(q,k)$, and some k-dimensional linear MDS code over K has length exactly $B(q,k)$.

Theorem 1.1 resolves the MDS conjecture for every field of even order. The codes of length $B(q,k)$ are classical (Section 2); the new content is the upper bound for $4 \le k \le q-2$, which was previously known only in the restricted ranges listed above. In particular, over $\mathbb{F}_{256}$ no linear MDS code of dimension $4 \le k \le 254$ is longer than 257, so the extended RS codes used in practice already have the largest possible length. The theorem concerns lengths of *linear* codes; it does not classify the codes attaining the bound, and fields of odd order lie outside its scope.

The proof turns the problem into one about plane arcs (Section 3). A hypothetical long arc in K^r produces one plane arc together with $t = r-3$ ways of "lifting" it into four-dimensional space, and these lifts together with three coordinate vectors give $t+3$ independent vectors. We show that all lifts of a plane arc lie in a single vector space of dimension at most $t+2$, which is impossible. Figure 2 shows the structure of the argument.

This report makes four contributions: (1) a proof of the upper bound in Theorem 1.1, which resolves the MDS conjecture for all fields of even order; (2) the common-height method behind it, which controls all lifts

of a plane arc at once rather than one at a time (Sections 3–7); (3) a complete formal proof, verified in the Lean proof assistant (Section 8); and (4) computer programs that test every lemma on 250 explicit examples and settle small fields by exhaustive search (Section 8).

2 Preliminaries

2.1 Arithmetic in characteristic two

Throughout, $K = \mathbb{F}_q$ with $q = 2^e$. We use the following standard facts [44].

Lemma 2.1. *(a)* $1 + 1 = 0$*, so* $-x = x$*. Hence determinants do not change when two rows are swapped, and the cross product* $a \times b$ *needs no signs.*

(b) $(x + y)^2 = x^2 + y^2$*, and therefore* $(x + y)^{2^m} = x^{2^m} + y^{2^m}$*, for field elements and for polynomials.*

(c) $x^q = x$ *for all* $x \in K$*. Hence squaring is a bijection of* K*, and every element has exactly one square root, namely* $x^{q/2}$*.*

(d) A polynomial is the square of a polynomial if and only if every exponent of every monomial in it is even. Its square root is obtained by halving the exponents and taking square roots of the coefficients.

(e) $\partial(X^m)/\partial X = mX^{m-1}$ *vanishes when* m *is even. In particular derivatives kill squares:* $\partial_i(G^2) = 2G\,\partial_i G = 0$*.*

Here ∂_i denotes the partial derivative with respect to X_i.

2.2 Polynomials, lines and root counting

Let $R = K[X_0, X_1, X_2]$ and let R_d be the space of homogeneous polynomials (forms) of degree d; $\dim R_d = \binom{d+2}{2}$ and $R_d = 0$ for $d < 0$. A nonzero $a \in K^3$ defines the linear form $\ell_a(X) = a \cdot X$ and the projective line $L_a = \{\ell_a = 0\}$. All identities between polynomials are identities of coefficients, not merely of functions on K.

Lemma 2.2 (Root counting [44, 18])**.** *(a) A nonzero form in two variables of degree* d *vanishes at no more than* d *points of the projective line.*

(b) A nonzero linear form ℓ *divides* $f \in R$ *if and only if the restriction of* f *to the plane* $\{\ell = 0\}$ *is the zero polynomial.*

(c) If pairwise non-proportional linear forms all divide f*, then so does their product.*

Proof. (a) Each zero $(u : v)$ contributes a factor $vU + uV$; distinct zeros give non-proportional factors, and their degrees add up. (b) After renaming variables and scaling, $\ell = X_2 + \alpha X_0 + \beta X_1$. Dividing f by ℓ as a polynomial in X_2 leaves the remainder $f(X_0, X_1, \alpha X_0 + \beta X_1)$, which is the restriction. (c) If $\ell_1 \mid f$, write $f = \ell_1 g$; restricting to $\{\ell_2 = 0\}$, where ℓ_1 is a nonzero form, shows $\ell_2 \mid g$. Induct. □

2.3 Codes and arcs

A linear $[N, k]$ code is a k-dimensional subspace of K^N. The *weight* of a word is its number of nonzero entries; the minimum distance d of a code is the least weight of a nonzero codeword; the code is MDS if $d = N - k + 1$. A *rank-k arc* of length $N \geq k$ is a list of N vectors in K^k, every k of which are linearly independent. A rank-3 arc is a *plane arc*.

Lemma 2.3 (Generator criterion [47])**.** *Let G be a $k \times N$ matrix of rank k. Its row space is MDS if and only if the columns of G form a rank-k arc.*

Proof. A codeword is xG with $x \in K^k$. If every k columns are independent, xG cannot vanish in k positions unless $x = 0$, so nonzero words have weight $\geq N - k + 1$; a nonzero word vanishing on any $k - 1$ chosen positions exists by counting dimensions, so the minimum is exactly $N - k + 1$. Conversely, if k columns are dependent, some $x \neq 0$ is orthogonal to all of them and xG is a nonzero word of weight $\leq N - k$. □

Lemma 2.4 (Operations on arcs [47, 33])**.** *Let $a_1, \ldots, a_N$ be a rank-k arc.*

(a) Any $M \geq k$ of the vectors form a rank-k arc.

(b) If $k \geq 2$, the images of $a_1, \ldots, a_{N-1}$ in $K^k/\langle a_N \rangle \cong K^{k-1}$ form a rank-$(k-1)$ arc.

(c) (Duality.) There is a rank-$(N-k)$ arc of length N.

Proof. (a) is clear. (b) A dependence among $k - 1$ images lifts to a dependence among those vectors and a_N. (c) Let $D \subseteq K^N$ be the space of coefficient vectors x with $\sum x_i a_i = 0$; $\dim D = N - k$. A vector of D vanishing on $N - k$ chosen positions gives a relation among the remaining k vectors, so it is zero. Thus restriction to any $N - k$ positions is an isomorphism, and the columns of a basis matrix of D form the arc. □

Lemma 2.5 (Small ranks [10])**.** *A rank-2 arc has length at most $q + 1$, and a rank-3 arc has length at most $q + 2$.*

Proof. A rank-2 arc consists of pairwise non-proportional vectors, and K^2 has exactly $q + 1$ lines through the origin. For rank 3 apply Lemma 2.4(b). □

Proposition 2.6 (Classical constructions [56, 10, 33])**.** *(a) Let $1 \leq k \leq q + 1$, and for $x \in K$ put $v_x = (1, x, x^2, \ldots, x^{k-1})$. The q vectors v_x together with $(0, \ldots, 0, 1)$ form a rank-k arc of length $q + 1$ (the extended Reed–Solomon code).*

(b) The vectors $(1, x, x^2)$, $x \in K$, together with $(0, 0, 1)$ and $(0, 1, 0)$ form a plane arc of length $q + 2$ (the regular hyperoval). By duality there is also a rank-$(q-1)$ arc of length $q + 2$.

Proof. (a) A $k \times k$ determinant of such columns is a Vandermonde determinant, or, if $(0, \ldots, 0, 1)$ is used, a Vandermonde determinant of size $k - 1$; both are nonzero for distinct x. (b) Three finite columns

give a Vandermonde determinant. Two finite columns with $(0,0,1)$ give $y-x$, and with $(0,1,0)$ give $y^2-x^2=(y-x)^2$ by Lemma 2.1(b). One finite column with both extra columns gives 1. □

Note that (b) uses characteristic two: for odd q, y^2-x^2 vanishes when $y=-x$.

3 Reduction to one statement about plane arcs

From now on $q=2n$ with $n\geq 2$; note that $n=q/2$ is a power of two.

Theorem 3.1 (Key Theorem)**.** *If $4\leq r\leq n+1$, there is no rank-r arc of length $q+2$ over K.*

Proof of Theorem 1.1 from Theorem 3.1. The constructions of Proposition 2.6 have length $B(q,k)$. For the upper bound, first let $k\notin\{3,q-1\}$ and suppose a rank-k arc of length at least $q+2$ exists. Keep $q+2$ of its vectors. By duality there is also a rank-$(q+2-k)$ arc of length $q+2$, so there is one of rank $r=\min\{k,q+2-k\}$, where $2\leq r\leq n+1$. The case $r=2$ contradicts Lemma 2.5; $r=3$ would force $k\in\{3,q-1\}$; and $4\leq r\leq n+1$ contradicts Theorem 3.1. For $k=3$ use Lemma 2.5. Finally let $k=q-1\neq 3$, so $q\geq 8$. A rank-$(q-1)$ arc of length $q+3$ would have a dual of rank 4 and length $q+3$; deleting one vector contradicts Theorem 3.1 with $r=4\leq n+1$. The code statement follows from Lemma 2.3. □

The coordinate trick. Suppose a rank-r arc of length $q+2$ exists, with $4\leq r\leq n+1$. Put

$$t=r-3,\qquad M=q+2-t,\qquad \text{so}\qquad 1\leq t\leq n-2,\quad M\geq r.$$

Choose t of the vectors as *centers* and three more vectors. These r vectors form a basis; in the coordinates of this basis the three vectors become e_1,e_2,e_3 and the centers become $e_4,\ldots,e_r$. The M vectors other than the centers can then be written

$$v_i=(a_i,\ h_i^{(1)},\ \ldots,\ h_i^{(t)})\qquad (a_i\in K^3,\ 1\leq i\leq M).$$

(i) The a_i form a plane arc: three v_i together with the t centers are r independent vectors, and their determinant equals the 3×3 determinant of the corresponding a_i.

(ii) For each j, the vectors e_4 and $(a_i,h_i^{(j)})$, $1\leq i\leq M$, form a rank-4 arc: four of them, together with the other $t-1$ centers, are r independent vectors, and the determinant reduces to the 4×4 one.

(iii) The $r\times M$ matrix with columns v_i has rank r, because $M\geq r$ of its columns are arc vectors. Hence its $r=t+3$ rows are linearly independent vectors of K^M.

Definition 3.2. Let $a_1, \dots, a_M$ be a plane arc. A *lift* is a vector $h \in K^M$ for which $e_4 = (0,0,0,1)$ and the M vectors $(a_i, h_i) \in K^4$ form a rank-4 arc. A *shear* is a vector $h(s) = (a_i \cdot s)_{i=1}^{M}$ with $s \in K^3$.

The first three rows of the matrix in (iii) are the coordinate rows of the a_i, that is, the shears $h(s)$ for $s = (1,0,0), (0,1,0), (0,0,1)$; by (ii) the other t rows are lifts. The Key Theorem therefore follows from the next result, whose proof occupies Sections 4–7.

Theorem 3.3 (Capacity Theorem)**.** *Let $1 \le t \le n-2$, $M = q+2-t$, and let $a_1, \dots, a_M$ be a plane arc. There is a linear subspace $\mathcal{H} \subseteq K^M$ of dimension at most $t+2$ that contains every shear and every lift.*

Proof of Theorem 3.1 from Theorem 3.3. By (iii) the three coordinate shears and the t lifts are $t+3$ independent vectors, but they all lie in $\mathcal{H}$, whose dimension is at most $t+2$. □

The margin is a single dimension, and the computations of Section 8 show that the bound $t+2$ is attained by many plane arcs. The capacity theorem cannot be proved by looking at one lift at a time: its force is that all lifts fit into one common space.

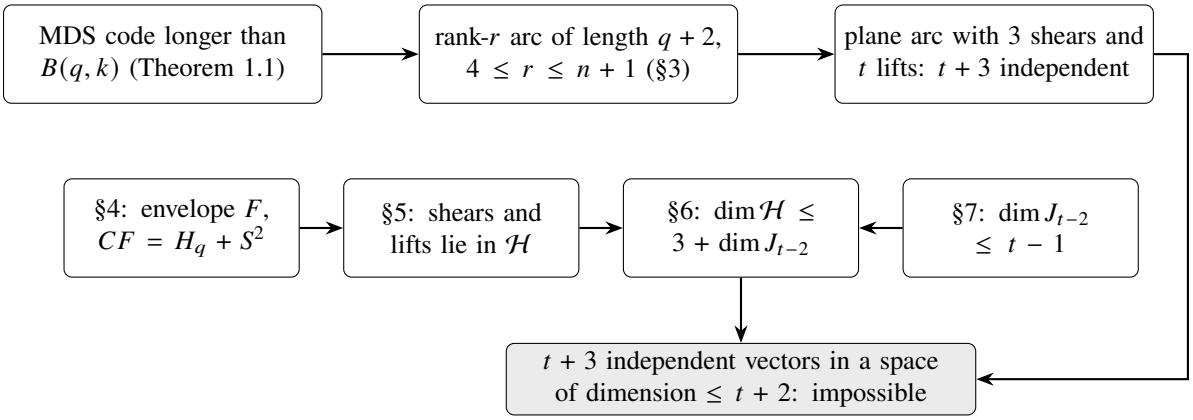


Figure 2: Structure of the proof. Sections 4–7 prove the Capacity Theorem 3.3, which contradicts the $t+3$ independent vectors produced from a hypothetical long arc.

Image created by the student researcher using TikZ [70], 2026.

Notation for the rest of the proof. Fix $1 \le t \le n-2$, $M = q+2-t$ and a plane arc $a_1, \dots, a_M$. Put

$$\ell_i = \ell_{a_i}, \qquad C = \prod_{i=1}^{M} \ell_i, \qquad C_i = \prod_{j \ne i} \ell_j, \qquad p_{ij} = a_i \times a_j, \qquad c_{ij} = \prod_{k \ne i,j} \ell_k(p_{ij}).$$

The vector p_{ij}, called a *node*, represents the point where the lines $L_i = L_{a_i}$ and L_j meet (Figure 3).

Lemma 3.4 (Geometry of the nodes)**.** *(a) $\ell_k(p_{ij}) = \det(a_i, a_j, a_k) \ne 0$ for $k \notin \{i,j\}$; so $c_{ij} \ne 0$ and no three of the lines L_i pass through one point.*

(b) For fixed i, the $M-1$ nodes p_{ij} $(j \neq i)$ are distinct points of L_i.

(c) (Star vanishing.) *A form of degree $d < M-1$ that vanishes at every node is zero.*

Proof. (a) is the triple-product formula. (b) If p_{ij} and p_{ik} were proportional, $\ell_k(p_{ij})$ would vanish. (c) On each line L_i the form has $M-1 > d$ zeros, so by Lemma 2.2 it is divisible by every ℓ_i, hence by C, whose degree M exceeds d. □

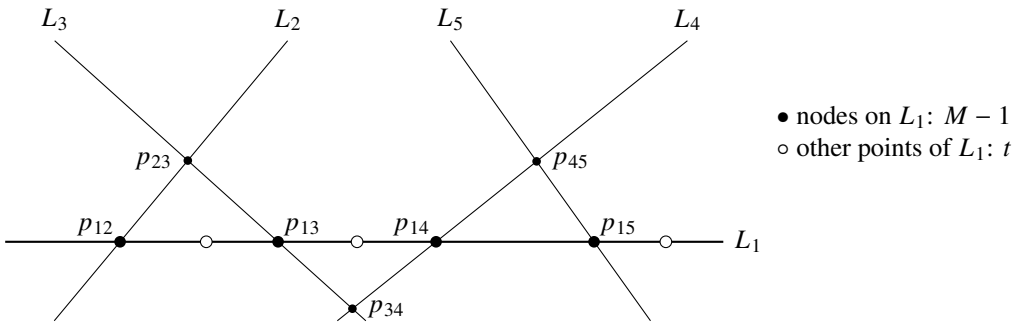


Figure 3: Schematic picture of the lines $L_i = \{a_i \cdot X = 0\}$ of a plane arc (drawn over the real numbers for illustration). Each line contains $M-1$ nodes p_{ij}, and no three lines meet. Since a line over K has $q+1$ points, each line has exactly $q+2-M = t$ further points; these correspond to the tangent lines of the arc [62, 33].

Image created by the student researcher using TikZ [70], 2026.

4 The normalized envelope

Idea. On each line L_i the $M-1$ nodes leave t other points (Figure 3). The product of the t linear forms vanishing at those points is a polynomial T_i of degree t on the line. We glue the M local polynomials T_i into one plane polynomial F; the gluing works because every T_i comes with an exact scaling, and the scalings agree at the nodes. Then a parity argument turns this exact scaling into the identity $CF = H_q + S^2$, which drives the rest of the proof.

Segre showed that for even q the tangent lines of a plane arc of size $q+2-t$ lie on an algebraic curve of class t (an "envelope") [61, 62, 33]; his lemma of tangents goes back to [60]. We need a version in which the polynomial is normalized exactly, in the spirit of Ball's coordinate-free form of the lemma of tangents [3, 4].

Definition 4.1. A *normalized envelope* is a form $F \in R_t$ with $c_{ij}\, F(p_{ij}) = 1$ for all $i \neq j$.

Everything rests on the polynomial $\Pi(U,V) = U^q V + UV^q$. Every nonzero linear form $\lambda = \alpha U + \beta V$ divides Π, because λ vanishes at (β, α) and $\Pi(\beta,\alpha) = \beta\alpha + \beta\alpha = 0$ by Lemma 2.1(c). Thus Π is a constant times the product of $q+1$ linear forms, one vanishing at each point of the projective line, and $\nabla\Pi = (V^q, U^q)$.

Lemma 4.2 (Lemma of tangents on one line)**.** *Let $\lambda_1, \dots, \lambda_m$ be pairwise non-proportional linear forms in U, V with $m + t = q + 1$. Suppose there are nonzero $w_j \in K^2$ and one constant $\kappa \neq 0$ such that*

$\lambda_j(U,V) = \kappa(w_{j2}U + w_{j1}V)$ *for every* j *(so* $\lambda_j(w_j) = 0$*). Then there is a form* $T(U,V)$ *of degree* t *with* $\big(\prod_{l\neq j}\lambda_l(w_j)\big)T(w_j) = 1$ *for every* j.

Proof. By Lemma 2.2(c), $\prod_l \lambda_l$ divides Π; define T by $\big(\prod_l \lambda_l\big)T = \kappa\Pi$. Take the gradient of both sides at w_j. On the left only the term in which λ_j is differentiated survives, giving $\big(\prod_{l\neq j}\lambda_l(w_j)\big)T(w_j)\,\kappa(w_{j2}, w_{j1})$. On the right we get $\kappa(w_{j2}^q, w_{j1}^q) = \kappa(w_{j2}, w_{j1})$. Since $w_j \neq 0$, the scalar factors agree. □

To use this on the line L_i, pick two other labels b, c, put $\Delta = \det(a_i, a_b, a_c) \neq 0$, and parametrize the plane $\{\ell_i = 0\}$ by $\phi_i(U,V) = Up_{ib} + Vp_{ic}$. For $j \neq i$, the restricted form is

$$\lambda_j(U,V) = \ell_j(\phi_i(U,V)) = U\det(a_i, a_b, a_j) + V\det(a_i, a_c, a_j),$$

and $p_{ij} = \phi_i(w_j)$ with $w_j = \Delta^{-1}\big(\det(a_i, a_j, a_c), \det(a_i, a_j, a_b)\big)$, as one checks by taking dot products with a_b and a_c. Because swapping two rows does not change a determinant (Lemma 2.1(a)), the coefficient vector of λ_j is exactly $\Delta(w_{j2}, w_{j1})$: the hypothesis of Lemma 4.2 holds with $\kappa = \Delta$, the *same* constant for every j. Since $\prod_{l\neq i,j}\lambda_l(w_j) = c_{ij}$, we obtain a form T_i of degree t with

$$c_{ij}\,T_i(w_j) = 1 \qquad (j \neq i), \qquad \text{where } p_{ij} = \phi_i(w_j). \tag{4.1}$$

Theorem 4.3 (Normalized envelope)**.** *There is exactly one normalized envelope* F.

Proof. Interpolation. Since $2t \leq q$, we have $M \geq t + 2$. Let E be a set of $t + 2$ labels and, for each pair $S = \{i, j\} \subset E$, let $B_S = \prod_{k\in E\setminus S}\ell_k \in R_t$. Put

$$F = \sum_{S=\{i,j\}\subset E} \frac{B_S}{c_{ij}\,B_S(p_{ij})}.$$

At a node p_{ij} with $i, j \in E$, every $B_{S'}$ with $S' \neq \{i, j\}$ contains ℓ_i or ℓ_j and vanishes, so $c_{ij}F(p_{ij}) = 1$.

Propagation. If $c_{ij}F(p_{ij}) = 1$ holds for more than t labels j on one line L_i, then the binary forms $F \circ \phi_i$ and T_i of degree t agree at more than t distinct points by (4.1), so they are equal and the normalization holds for every $j \neq i$. For $i \in E$, the $t + 1$ labels in $E \setminus \{i\}$ start this propagation, so every pair meeting E is normalized. For $i \notin E$, the $t + 2$ pairs $\{i, j\}$ with $j \in E$ are now normalized, and propagation finishes all pairs.

Uniqueness. The difference of two envelopes vanishes at all nodes and has degree $t < M - 1$, so it is zero by Lemma 3.4(c). □

The next identity is where squares enter. Put

$$H_q = X_0^q X_1 X_2 + X_0 X_1^q X_2 + X_0 X_1 X_2^q.$$

Theorem 4.4 (Highest identity)**.** *There is $S \in R_{n+1}$ with $CF = H_q + S^2$. Moreover $\ell_i \nmid F$ for every i.*

Proof. CF has even degree $M + t = 2n + 2$, so each of its monomials has zero or two odd exponents. Sorting monomials by which exponents are odd and using Lemma 2.1(d),

$$CF = X_1X_2A_0^2 + X_0X_2A_1^2 + X_0X_1A_2^2 + S^2, \qquad A_i \in R_n,\ S \in R_{n+1}.$$

Let $D_0 = \partial_1\partial_2$. By Lemma 2.1(e), D_0 kills every term except the first, so $D_0(CF) = A_0^2$. Now evaluate at a node, writing $CF = \ell_u\ell_v \cdot (C_{uv}F)$ with $C_{uv} = C/(\ell_u\ell_v)$. At p_{uv} both ℓ_u and ℓ_v vanish, so in the second derivative only the terms differentiating each of them once survive:

$$A_0(p_{uv})^2 = (a_{u1}a_{v2} + a_{u2}a_{v1})\, c_{uv}F(p_{uv}) = (p_{uv})_0 = \big((p_{uv})_0^n\big)^2,$$

using the normalization and $x = x^q = (x^n)^2$. Taking square roots, the form $A_0 - X_0^n$ of degree n vanishes at every node. Since $n < M - 1 = 2n + 1 - t$, Lemma 3.4(c) gives $A_0 = X_0^n$; likewise $A_1 = X_1^n$ and $A_2 = X_2^n$, which is the identity. Finally $F(p_{ij}) \neq 0$ at the nodes on L_i, so $\ell_i \nmid F$. □

Example. Over $\mathbb{F}_8$ take the $M = 8$ points $a_s = (1, s, s^2)$, $s \in \mathbb{F}_8$, so $t = 2$. Our program solves the 28 equations $c_{ij}F(p_{ij}) = 1$ and finds the unique solution $F = X_1X_2$, while

$$C = \prod_{s\in\mathbb{F}_8} (X_0 + sX_1 + s^2X_2) = X_0^8 + X_0X_1^7 + X_0X_2^7 + X_0^2X_1^5X_2 + X_0^4X_1X_2^3.$$

One checks by hand that $CF = H_8 + (X_0^2X_1X_2^2 + X_0X_1^3X_2)^2$, as Theorem 4.4 predicts. Geometrically, a point X of the plane of Figure 3 stands for the line $\{Y : X \cdot Y = 0\}$ of the original plane, and $F = X_1X_2$ vanishes exactly on the lines through $(0, 1, 0)$ or $(0, 0, 1)$: the two tangent lines at each point a_s.

5 Every lift is compatible

Idea. A lift is itself an arc in K^4, so it has its own envelope $G(X, z)$. Setting $z = 0$ gives back F, and the coefficient of z gives a polynomial Q. The four-dimensional highest identity then forces the pair (h, Q) to satisfy one fixed system of linear equations. Every lift therefore lies in the same vector space $\mathcal{H}$, defined below.

For $h \in K^M$ and $Q \in R_{t-1}$ define

$$P(h, Q) = CQ + F\sum_{i=1}^{M} h_iC_i \ \in R_{q+1}, \qquad D = \partial_0\partial_1\partial_2,$$

$$\mathcal{V} = \{(h, Q) :\ D\,P(h, Q) = 0\}, \qquad \mathcal{H} = \{h :\ (h, Q) \in \mathcal{V} \text{ for some } Q\}.$$

These are vector spaces, because the condition is linear. We show that $\mathcal{H}$ contains every shear and every lift.

Shears. For $s \in K^3$ write $\partial_s = s_0\partial_0 + s_1\partial_1 + s_2\partial_2$. The product rule gives $\partial_s C = \sum_i (a_i \cdot s)C_i$, hence

$$P(h(s), \partial_s F) = \partial_s(CF) = \partial_s H_q, \tag{5.1}$$

because $\partial_s(S^2) = 0$. Each term of H_q contains an even power X_i^q, so $DH_q = 0$, and $D\partial_s H_q = \partial_s DH_q = 0$. Thus $(h(s), \partial_s F) \in \mathcal{V}$.

Lifts. Here we need the same envelope theory one dimension higher. For a rank-4 arc $b_1, \dots, b_A$ in K^4 with $A + t = q + 3$, the node of a triple $S = \{i, j, k\}$ is the vector p_S defined by $p_S \cdot v = \det(b_i, b_j, b_k, v)$, and $c_S = \prod_{l \notin S} \ell_{b_l}(p_S)$.

Theorem 5.1 (Space envelope)**.** *There is exactly one form $G \in K[X_0, X_1, X_2, X_3]_t$ with $c_S G(p_S) = 1$ for every triple S. Moreover, $P_4 = \big(\prod_l \ell_{b_l}\big)G$ satisfies $\big(\prod_{j\neq i} \partial_j\big)P_4 = X_i^q$ for $i = 0, 1, 2, 3$.*

Proof. The proofs of Theorems 4.3 and 4.4 carry over with these changes. (1) For fixed i, j, the nodes p_{ijk} lie on the plane $\{\ell_{b_i} = \ell_{b_j} = 0\}$, which we parametrize by $Up_{ijc} + Vp_{ijd}$; the same determinant-swap computation gives Lemma 4.2 with $\kappa = \det(b_i, b_j, b_c, b_d)$ and $m = A - 2$. (2) Interpolate on a set E of $t + 3 \le A$ labels with products over $E \setminus S$ for triples S. (3) Propagate in three passes: from triples with three labels in E to those with two, then one, then none; each pass starts from at least $t + 1$ normalized nodes. (4) For uniqueness, restrict to each space $\{\ell_{b_i} = 0\}$; the other $A - 1$ forms restrict to linear forms in three variables, any three of them independent (because any four b_l are), and their nodes are the p_{ijk}. Star vanishing (Lemma 3.4(c), which holds for any plane arc) applies because $t < A - 2$. (5) P_4 has odd degree $2n + 3$, so each monomial has one or three odd exponents, and

$$P_4 = \sum_{i=0}^{3} \Big(\prod_{j\neq i} X_j\Big) A_i^2 + \sum_{i=0}^{3} X_i B_i^2.$$

The third derivative $\prod_{j\neq i} \partial_j$ maps P_4 to A_i^2. At a node p_S only the terms differentiating the three vanishing factors once each survive; their sum is the 3×3 minor $(p_S)_i$ (signs do not matter), times $c_S G(p_S) = 1$. So $A_i(p_S)^2 = (p_S)_i$, and as before $A_i = X_i^n$, since $n < A - 2$. □

Theorem 5.2 (Compatibility)**.** *Every lift h lies in $\mathcal{H}$.*

Proof. Write z for the fourth variable. The lifted arc $e_4, (a_1, h_1), \dots, (a_M, h_M)$ has $A = M + 1$ vectors, so $A + t = q + 3$ and Theorem 5.1 gives its envelope $G(X, z)$. The product of its linear forms is zB with $B = \prod_i (\ell_i + h_i z)$.

Restriction. The node of $e_4, (a_i, h_i), (a_j, h_j)$ is $(p_{ij}, 0)$ (expand the determinant along e_4), and the other factors evaluate there to c_{ij}. Thus $G(X, 0)$ is a normalized plane envelope, and $G(X, 0) = F$ by uniqueness.

Derivative. Theorem 5.1 with $i = 3$ says $D(zBG) = z^q$. Since D does not involve z, we get $D(BG) = z^{q-1}$. Differentiate in z and set $z = 0$; as $q \geq 4$, the right side becomes 0. With $Q = (\partial_z G)(X, 0) \in R_{t-1}$ and the product rule,

$$\partial_z(BG)\big|_{z=0} = F \sum_i h_i C_i + CQ = P(h, Q),$$

so $D\,P(h, Q) = 0$. □

6 The Frobenius obstruction

Idea. It remains to bound $\dim \mathcal{H}$. We attach to each compatible pair a polynomial Φ of degree $2n$ by a square-root trick. The pairs with $\Phi = 0$ form a space of dimension at most 3, and for the other pairs $\Phi = Cg$ with g in a small space J_{t-2}. Counting dimensions gives $\dim \mathcal{H} \leq 3 + \dim J_{t-2}$.

Call a polynomial *n-heavy* if every monomial in it has some exponent at least n, and let

$$J_d = \{f \in R_d : \ Cf \text{ is } n\text{-heavy}\}.$$

(In algebraic language J is the colon ideal $(X_0^n, X_1^n, X_2^n) : C$ [18, 23].) We use the operator $\Theta = X_0^q \partial_0 + X_1^q \partial_1 + X_2^q \partial_2$, which obeys the product rule and satisfies $\Theta \ell = \ell^q$ for every linear form ℓ with coefficients in K, by Lemma 2.1(b),(c).

Lemma 6.1 (Parity form)**.** *If $(h, Q) \in \mathcal{V}$ and $P = P(h, Q)$, then $P = X_0 B_0^2 + X_1 B_1^2 + X_2 B_2^2$ with $B_j \in R_n$, and $\Theta P = \Phi^2$ where $\Phi = X_0^n B_0 + X_1^n B_1 + X_2^n B_2$.*

Proof. P has odd degree $2n + 1$, so each monomial has one or three odd exponents. D sends a monomial to a nonzero multiple of another monomial exactly when all three exponents are odd, and distinct monomials go to distinct monomials; so $DP = 0$ means every monomial of P has exactly one odd exponent. Grouping by the odd variable gives the form of P. Then $\partial_j P = B_j^2$ and $\Theta P = \sum_j X_j^{2n} B_j^2 = \Phi^2$. □

Lemma 6.2 (Divisibility)**.** *In Lemma 6.1, C divides Φ, and $\Phi = Cg$ with $g \in J_{t-2}$.*

Proof. By Theorem 4.4, $\Theta(CF) = \Theta H_q + \Theta(S^2)$. Here $\Theta(S^2) = 0$, and a direct computation shows that each of the three monomials of ΘH_q appears twice, so $\Theta(CF) = 0$. Since $0 = \Theta(\ell_i C_i F) = \ell_i^q C_i F + \ell_i \Theta(C_i F)$, we get $\Theta(C_i F) = \ell_i^{q-1} C_i F$. Now reduce ΘP modulo ℓ_i: the terms CQ and $h_k C_k F$ with $k \neq i$ are multiples $\ell_i Y$, and $\Theta(\ell_i Y) = \ell_i^q Y + \ell_i \Theta Y$ is again a multiple of ℓ_i; the remaining term gives $h_i \ell_i^{q-1} C_i F$, also a multiple of ℓ_i. So $\ell_i \mid \Theta P = \Phi^2$. Restricting to L_i shows $\ell_i \mid \Phi$ (a square of a nonzero polynomial is nonzero), and by Lemma 2.2(c), $C \mid \Phi$. The quotient g has degree $2n - M = t - 2$, and $Cg = \Phi$ is n-heavy. □

Lemma 6.3 (Small kernel)**.** *The pairs $(h, Q) \in \mathcal{V}$ with $\Phi = 0$ form a space of dimension at most* 3*.*

Proof. The map $(h, Q) \mapsto P(h, Q)$ is injective. If $P = 0$, restricting to L_i kills every term except $h_i C_i F$; neither C_i nor F is divisible by ℓ_i (Theorem 4.4), so $h_i = 0$. Then $CQ = 0$ and $Q = 0$.

If $\Phi = 0$, then P lies in a fixed 3-dimensional space. Suppose $\sum_j X_j^n B_j = 0$. A monomial X^m of B_0 gives $X_0^n X^m$, which can only cancel against $X_1^n B_1$ or $X_2^n B_2$ if X^m is X_1^n or X_2^n. Arguing the same way for B_1, B_2 and comparing the coefficients of $X_0^n X_1^n$, $X_0^n X_2^n$, $X_1^n X_2^n$ gives constants α, β, γ with $B_0 = \alpha X_1^n + \beta X_2^n$, $B_1 = \alpha X_0^n + \gamma X_2^n$, $B_2 = \beta X_0^n + \gamma X_1^n$. Then $P = \gamma^2 \partial_0 H_q + \beta^2 \partial_1 H_q + \alpha^2 \partial_2 H_q$. □

Proposition 6.4. $\dim \mathcal{H} \le \dim \mathcal{V} \le 3 + \dim J_{t-2}$.

Proof. The first inequality holds because $\mathcal{H}$ is the image of $\mathcal{V}$ under $(h, Q) \mapsto h$. The map $\Lambda(h, Q) = \Theta P(h, Q)$ is linear. By Lemmas 6.1–6.3, its kernel has dimension at most 3 and its image lies in $C^2 \cdot \{g^2 : g \in J_{t-2}\}$. For any subspace W of polynomials, $\{g^2 : g \in W\}$ is a subspace of the same dimension: $\sum c_i g_i^2 = (\sum \sqrt{c_i}\, g_i)^2$ by Lemma 2.1(b),(c), so squares of a basis span it, and they are independent because a polynomial whose square is 0 is 0. Rank–nullity finishes the proof. □

When $t = 1$, C has degree $2n + 1$ and cannot divide the degree-$2n$ polynomial Φ unless $\Phi = 0$; so $\dim \mathcal{H} \le 3 = t + 2$ already. For $t \ge 2$ it remains to show $\dim J_{t-2} \le t - 1$.

7 Counting annihilators

Idea. Two facts combine. First, no nonzero polynomial of degree below $d = t - 2$ lies in J; this uses the lines L_i once more. Second, a certain sequence of matrix ranks is symmetric and grows until its middle. Together they leave room for at most $d + 1$ independent polynomials in J_d.

Put $d = t - 2$, so $M = 2n - d$ and $0 \le d \le n - 4$. If $d = 0$ then $\dim J_0 \le \dim R_0 = 1$ and we are done, so let $d \ge 1$. We first show that no nonzero form of degree below d is in J, using one more Frobenius trick.

Lemma 7.1 (No early annihilators)**.** $J_j = 0$ *for* $0 \le j < d$.

Proof. Let $f \in J_j$. Assign each monomial of the n-heavy polynomial Cf to a variable whose exponent is at least n: $Cf = X_0^n B_0 + X_1^n B_1 + X_2^n B_2$ with B_k of degree $r = n - d + j < n$. Fix a line L_i; after renaming variables its equation is $X_2 = \alpha X_0 + \beta X_1$. Because n is a power of two, Lemma 2.1(b) gives $X_2^n = \alpha^n X_0^n + \beta^n X_1^n$ on L_i, so

$$X_0^n \big(B_0 + \alpha^n B_2\big) + X_1^n \big(B_1 + \beta^n B_2\big) = 0 \quad \text{on } L_i.$$

The two brackets have degree $r < n$, so the two terms share no monomial and both brackets vanish. Hence at every point x of L_i the vector $(B_0(x), B_1(x), B_2(x))$ is a multiple of $(\alpha^n, \beta^n, 1)$, the coordinatewise n-th

power of the coefficient vector of the line. At a node, this holds for two lines; the n-th powers of two independent vectors are independent (each 2×2 minor is raised to the n-th power), so all B_k vanish at every node. As $r < M - 1$, star vanishing (Lemma 3.4(c)) gives $B_k = 0$, so $Cf = 0$ and $f = 0$. □

To count J_d itself we use a pairing. Let $N = 3(n-1)$ and call $X_0^{n-1}X_1^{n-1}X_2^{n-1}$ the *top monomial*. For any form G of degree at most N define $J(G)_j = \{u \in R_j : Gu \text{ is } n\text{-heavy}\}$, the number $s(G) = N - \deg G$, and the *rank sequence* $h_j(G) = \dim R_j - \dim J(G)_j$. Thus $J_j = J(C)_j$ and $s(C) = n + d - 3$.

Lemma 7.2 (Top-coefficient pairing)**.** *For $u \in R_j$, $u \in J(G)_j$ if and only if the coefficient of the top monomial in Guv is 0 for every $v \in R_{s(G)-j}$. Consequently $h_j(G) = h_{s(G)-j}(G)$, and $h_j(G) = 0$ for $j > s(G)$.*

Proof. If Gu is n-heavy, so is Guv, and the top monomial (all exponents $n-1$) does not occur. If Gu is not n-heavy, it contains a monomial X^m with all $m_k \le n-1$; the complementary monomial $v = X^{(n-1)-m}$ has degree $s(G) - j$, and X^m is the only monomial of Gu that v moves to the top, so the top coefficient of Guv is nonzero. Hence $h_j(G)$ is the rank of the matrix of top coefficients of Gmm', with m and m' running over the monomials of degrees j and $s(G) - j$. This matrix for j is the transpose of the one for $s(G) - j$, and row rank equals column rank. □

Lemma 7.3 (Two-variable growth; a case of Macaulay's theorem [46])**.** *Let W_j $(j \ge 0)$ be spaces of forms of degree j in two variables Y, Z with $YW_j + ZW_j \subseteq W_{j+1}$, and put $b_j = (j+1) - \dim W_j$. If $b_j < j + 1$, then $b_{j+1} \le b_j$, and the sequence b never increases again.*

Proof. If $W_j \neq 0$, choose $f \in W_j$ in which the power of Z is as large as possible. Then $Zf \in W_{j+1}$ contains a higher power of Z than anything in YW_j, so $\dim W_{j+1} \ge \dim YW_j + 1 = \dim W_j + 1$. Also $W_{j+1} \neq 0$, so the argument repeats. □

Lemma 7.4 (The rank sequence rises to the middle)**.** *If G is not n-heavy and $s = s(G)$, then $h_{j-1}(G) \le h_j(G)$ for $1 \le j \le s/2$.*

Proof. Induction on s; for $s \le 1$ there is nothing to prove. Let $s \ge 2$. G has a monomial X^m with all $m_k \le n-1$, and since its degree is below N, some $m_k < n-1$. Let $c = X_k$; then Gc contains $X^m X_k$ and is not n-heavy, and $s(Gc) = s - 1$. Note $J(Gc)_{j-1} = \{u : cu \in J(G)_j\}$. The map $R_{j-1} \to R_j/J(G)_j$, $u \mapsto cu$, has kernel $J(Gc)_{j-1}$, so by rank–nullity

$$h_j(G) = a_j + b_j, \qquad a_j = h_{j-1}(Gc)\ (a_0 = 0), \qquad b_j = \dim R_j \big/ \big(J(G)_j + cR_{j-1}\big).$$

Setting $X_k = 0$ identifies R_j/cR_{j-1} with the forms of degree j in the other two variables, and $b_j = (j+1) - \dim W_j$, where W_j is the image of $J(G)_j$; these W_j satisfy Lemma 7.3. By induction and Lemma 7.2 for Gc: $a_{j-1} \le a_j$ for $1 \le j \le s/2$, and $a_j = a_{s+1-j}$ for $1 \le j \le s$.

Suppose $h_j(G) < h_{j-1}(G)$ for some $1 \le j \le s/2$. Then $b_j < b_{j-1} \le j$, so by Lemma 7.3 the sequence b is non-increasing from index j on; in particular $b_{s+1-j} \le b_j$. Using the symmetry of $h(G)$,

$$h_{j-1}(G) = h_{s+1-j}(G) = a_{s+1-j} + b_{s+1-j} \le a_j + b_j = h_j(G),$$

a contradiction. □

In commutative-algebra language, Lemmas 7.2 and 7.4 are a case of the fact that the Hilbert function of an Artinian Gorenstein quotient of $K[X_0, X_1, X_2]$ is symmetric and unimodal, a consequence of the Buchsbaum–Eisenbud structure theorem [66, 12, 38]. The induction above proves the case needed here directly, without the Lefschetz properties, which can fail in positive characteristic [50, 31].

Theorem 7.5 (Annihilator bound)**.** $\dim J_d \le d + 1$.

Proof. For $d \ge 1$, Lemma 7.1 with $j = 0$ shows that C is not n-heavy, and with $j = d - 1$ gives $h_{d-1}(C) = \binom{d+1}{2}$. Since $d \le n-3$, we have $d \le s(C)/2 = (n+d-3)/2$, so Lemma 7.4 gives $h_d(C) \ge \binom{d+1}{2}$, and $\dim J_d = \binom{d+2}{2} - h_d(C) \le \binom{d+2}{2} - \binom{d+1}{2} = d + 1$. □

Proof of the Capacity Theorem 3.3. $\mathcal{H}$ contains all shears (by (5.1)) and all lifts (Theorem 5.2). For $t = 1$ its dimension is at most 3, as noted after Proposition 6.4. For $2 \le t \le n-2$, Proposition 6.4 and Theorem 7.5 with $d = t - 2$ give $\dim \mathcal{H} \le 3 + (t - 1) = t + 2$. □

This completes the proof of the Key Theorem and of Theorem 1.1.

8 Computer verification

We wrote Python programs [55, 32, 69] implementing exact arithmetic in $\mathbb{F}_{2^e}$ by logarithm tables and exact Gaussian elimination over these fields. No floating-point arithmetic is used, so every computed dimension is exact.

8.1 Exhaustive search over small fields

For $q \in \{2, 4, 8\}$ and every $2 \le k \le q$ we computed the largest arc in K^k by complete backtracking search. An arc with at least $k + 1$ vectors can be moved by a change of coordinates so that it contains $e_1, \dots, e_k$ and $(1, \dots, 1)$; every further vector then has the form $(1, x_2, \dots, x_k)$ with distinct $x_i \notin \{0, 1\}$. The search adds such vectors in increasing order, discards candidates that would create k dependent vectors, and stops a branch when it cannot beat the best arc found. Table 1 shows that the maximum equals $B(q, k)$ in all eleven

Table 1: Exhaustive search for the largest rank-k arc (equivalently, the longest k-dimensional MDS code) over $\mathbb{F}_q$. "Nodes" is the number of partial arcs examined.

q	2	4	4	4	8	8	8	8	8	8	8
k	2	2	3	4	2	3	4	5	6	7	8
maximum found	3	5	6	5	9	10	9	9	9	10	9
$B(q,k)$	3	5	6	5	9	10	9	9	9	10	9
nodes	1	3	3	1	7	112	439	1559	3719	3719	1

Table created by the student researcher using Python [55] (program `arc_search.py` [69]), 2026.

cases, including the exceptional dimensions $k = 3$ and $k = q - 1$. These values agree with earlier results [34, 41].

Exhaustive search grows very quickly: over $\mathbb{F}_{16}$ with $k = 4$ there are already 2184 candidate vectors. Classifying the ovals of the plane over $\mathbb{F}_{16}$ required a substantial computer search [45], and the classification of all arcs for $4 \le q \le 19$ relied on extensive computation [41]. No computer search can cover all even q, which is why a proof is needed.

8.2 Checking every lemma on explicit arcs

A second program follows the proof step by step. Given a plane arc over $\mathbb{F}_q$ and $t = q + 2 - M$, it solves the linear equations $c_{ij}F(p_{ij}) = 1$ for F, checks that $CF + H_q$ is a perfect square, computes $\mathcal{V}$ and $\mathcal{H}$ by Gaussian elimination, computes Φ and checks that $C \mid \Phi$ (by evaluating at all points of all lines L_i), computes the rank sequence $h_j(C)$ of Section 7, and, for arcs with a known lift, checks that the lift is a rank-4 arc lying in $\mathcal{H}$. For $q \le 16$ it also solves for the space envelope G and checks $G(X, 0) = F$ and the four derivative identities of Theorem 5.1. Test arcs were random subsets of the regular hyperoval and of a translation hyperoval [33], random arcs built greedily, and arcs with lifts coming from the space curves $(1, s, s^{2^h}, s^{2^h+1})$, $\gcd(h, e) = 1$, which are arcs by a theorem of Casse and Glynn [15]; each test arc was moved by a random invertible matrix.

Table 2: Lemma-by-lemma checks. "Bound attained" counts arcs with $\dim \mathcal{H} = t + 2 < M$.

	$\mathbb{F}_8$	$\mathbb{F}_{16}$	$\mathbb{F}_{32}$	$\mathbb{F}_{64}$
values of t tested	1–4	1–8	1–16	9 values
plane arcs tested	42	90	93	25
arcs with a lift (lift in $\mathcal{H}$ checked)	18	42	60	16
space envelope checks	18	42	–	–
failed checks	0	0	0	0
bound $\dim \mathcal{H} = t + 2$ attained	19	45	33	18

Table created by the student researcher using Python [55] (programs `verify_capacity.py` and `verify_q64.py` [69]), 2026.

Table 2 summarizes the results; over $\mathbb{F}_{64}$ the program computes the two critical values h_{d-1}, h_d of the

rank sequence and skips the slow divisibility test. Every check passed in every case. Two observations go beyond the proof. First, the bound of the Capacity Theorem is attained: for arcs contained in a conic, $\dim \mathcal{H} = t+2$ in every case tested, so the contradiction in Section 3 really does come down to one dimension. Second, the rank sequences are symmetric and rise to the middle exactly as Lemmas 7.2 and 7.4 predict (Figure 4).

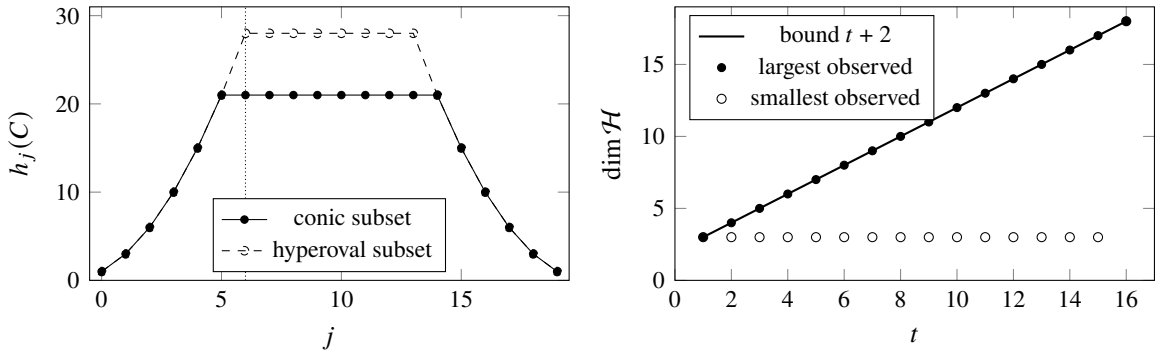


Figure 4: Left: rank sequences $h_j(C)$ for two plane arcs of size $M = 26$ over $\mathbb{F}_{32}$ ($t = 8$, $d = 6$, $s = 19$); both are symmetric and rise to the middle, and $\dim J_d = \binom{8}{2} - h_6$ equals $7 = d + 1$ for the conic subset and 0 for the other arc. Right: $\dim \mathcal{H}$ for all 93 test arcs over $\mathbb{F}_{32}$ against the bound $t + 2$ of the Capacity Theorem.

Graphs created by the student researcher using PGFPlots [24] and Python [55] (program `verify_capacity.py` [69]), 2026.

Because each lemma was checked separately, these computations test every step of the argument independently of the others.

8.3 Formal verification

The complete proof has been formalized and verified in the Lean 4 proof assistant [20] using the mathlib library [72], in 116 modules (about 11,000 lines) [68]. The final formal theorem states Theorem 1.1 for an arbitrary finite field of even cardinality, with codes as literal subspaces of K^N, dimension as vector space dimension, and minimum distance as the smallest Hamming weight of a nonzero codeword, so the formal statement matches Theorem 1.1 exactly. The formalization compiles with Lean 4.24.0 with no errors or warnings and contains no unproved steps, and the final theorem depends only on Lean's three standard axioms (`propext`, `Classical.choice` and `Quot.sound`). A version for current Lean (v4.35.0-rc2) and Mathlib states the theorem with Mathlib's own Hamming weight; the Comparator tool [43] confirms that its proof establishes exactly that statement and replays it through Lean's kernel and two independent kernel implementations (NanoDa and con-ron), all of which accept it. Formal verification has also settled other famous problems, such as the four colour theorem and the Kepler conjecture [2, 26, 29], and computer search proved the nonexistence of a projective plane of order 10 [42].

9 Discussion and conclusion

Where even order is used. Characteristic two enters in four places. The first is the square root of a polynomial (Lemma 2.1(d)), which creates the parity forms in Theorem 4.4 and Lemma 6.1. The others are the additivity of $x \mapsto x^n$ for $n = q/2$ (Lemma 7.1); the vanishing of all signs in the lemma of tangents (Section 4); and the existence of the hyperoval. For odd q several of these steps fail (squaring is no longer additive, and plane arcs have at most $q + 1$ points [10]), and large arcs can behave differently (for example, there is a non-classical 10-arc in $\mathrm{PG}(4, 9)$ [25]), so the method does not apply as it stands. The MDS conjecture for odd non-prime q remains open in general [7].

What is new. The constructions, the code–arc correspondence, Segre's envelope and the symmetry and unimodality of Gorenstein Hilbert functions are classical [47, 59, 61, 66, 33]. The new ingredients are the common height space $\mathcal{H}$, the proof that it contains every lift (Theorem 5.2), the Frobenius obstruction Φ that sends $\mathcal{H}$ modulo shears into J_{t-2}, and the geometric vanishing in Lemma 7.1 that makes the annihilator count possible. The argument belongs to the polynomial method in combinatorics [1, 22, 28, 71]: a geometric object is encoded by polynomials, and dimension counting finishes the proof. Table 3 lists the standard names of the objects used in this report.

Table 3: Terms used in this report and their standard names.

This report	Standard terminology	References
rank-k arc	arc in $\mathrm{PG}(k-1, q)$; columns of an MDS generator matrix	[33, 47]
normalized envelope F	Segre's envelope of class t of the tangents	[61, 62]
n-heavy polynomials	the ideal (X_0^n, X_1^n, X_2^n)	[18]
the spaces J_d	the colon ideal $(X_0^n, X_1^n, X_2^n) : C$	[18, 23]
rank sequence $h_j(C)$	Hilbert function of R/J	[66, 38]
top-coefficient pairing	Gorenstein (Macaulay) duality	[38, 23]
Lemma 7.4	unimodality of codimension-3 Gorenstein h-vectors	[66, 12]

Table created by the student researcher using LaTeX, 2026.

Scope. The theorem concerns linear codes over fields of even order. It determines the maximum length but does not classify the codes that attain it, and nonlinear codes and fields of odd order are separate questions.

Conclusion. This report resolves the linear MDS conjecture over every finite field of even order: for every dimension $2 \le k \le q$, the longest linear MDS code has length $q + 1$, except for length $q + 2$ in dimensions 3 and $q - 1$. This settles, in characteristic two, a problem that has stood since Segre's work of 1955 [59]. In particular, for the byte field $\mathbb{F}_{256}$ used in storage and cryptography, extended Reed–Solomon codes already

have the largest possible length. Natural next steps are to extend the common-height method to fields of odd order and to classify the arcs of maximum size in even characteristic.

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